Design Document: RL-Driven Physical Activity Interventions

A Configurable Simulation Framework

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1 Introduction

Reinforcement learning offers a principled approach to personalising just-in-time adaptive interventions (JITAIs) for physical activity, but training RL agents requires either expensive real-world trials or a realistic simulation environment. Existing simulators are bespoke and tied to specific datasets or papers, making cross-paper comparison difficult and limiting reproducibility. We present `rl-health-interventions`, a configurable simulation framework where the dataset schema, MDP dynamics, user behaviour models, and agent configurations are specified entirely through YAML config files—no source code changes required. The framework enables systematic comparison of RL agents on PA intervention tasks using synthetic data in Phase 1, with real-data integration planned for Phase 2.

The initial implementation delivers a rule-based user simulator with four behavioural archetypes, a multi-timescale MDP environment (daily steps + delayed 3-week clinical measures), and Thompson Sampling as the baseline agent [1, 2]. Deep RL agents (DQN, PPO) will be added if time permits. All components share a common ABC + registry interface for extensibility.

2 Literature Review

Reinforcement learning has been applied to health interventions through micro-randomized trials (MRTs), most notably HeartSteps V1 and V2, which delivered contextually-tailored activity suggestions and demonstrated the feasibility of RL-in-the-loop policy learning [2, 3]. These trials provide the only available source of real action-to-response mappings for calibrating user behaviour models.

Existing simulation environments for RL-driven health interventions tend to be tied to specific datasets or tasks. StepCountJITAI [1] provides a simulation environment for physical activity JITAIs but is scoped to a single dataset and action set. Health Gym [4] offers synthetic health environments but does not support configurable MDP specifications. Open wearable benchmark datasets such as ExtraSensory [5] and WISDM [6] are valuable for pipeline testing but lack intervention response data.

The gap we address is a config-first simulation framework where the dataset schema, MDP dynamics, user behaviour, and agent configurations are specified entirely through config files, enabling systematic cross-dataset and cross-policy comparisons without bespoke engineering.

3 MDP Formalisation

This section specifies the Markov Decision Process (MDP) for the first iteration of the simulation framework. The MDP is defined as the tuple $(\mathcal{S}, \mathcal{A}, P, R, \gamma)$.

State. The state represents the current context of a user at decision epoch t. States are constructed from wearable-device signals, user profile information, and intervention history:

$$s_t = \{\mathbf{x}_t^{\text{wearable}}, \mathbf{x}_t^{\text{context}}, \mathbf{x}_t^{\text{history}}, \mathbf{x}^{\text{profile}}\}$$

See Appendix A for the full list of state variables. Variables are normalised to $[0, 1]$ before being passed to the agent. The body measure is only observed every 3 weeks (sparse reward signal).

Actions.

$$\mathcal{A} = \{a_0, a_1, a_2, a_3, a_4, a_5\}$$

Table 1: Action space

Action	Label	Description
a_0	No message	Do not intervene
a_1	Motivational prompt	Generic encouragement
a_2	Walking suggestion	Recommend a short walk
a_3	Goal reminder	Remind of step goal
a_4	Recovery suggestion	Stretch or rest
a_5	Progress feedback	Report goal progress

Exactly one action per decision epoch. The "no message" action is critical for managing notification burden.

Transition. The transition function models the probability over next states:

$$P(s_{t+1} \mid s_t, a_t) : \mathcal{S} \times \mathcal{A} \times \mathcal{S} \to [0, 1]$$

In Phase 1 (rule-based), transitions follow a behavioural response model:

$$\Delta \text{steps}_{t+1} = f(s_t, a_t, \theta_{\text{user}})$$

where θ_{user} parameterises one of four behavioural archetypes:

- **Goal-driven:** Responds to reminders and feedback. Proportional to goal progress.
- **Social responder:** Responds to motivational prompts. Models social desirability bias.
- **Resistant:** Low response. Fatigue accumulates quickly.
- **Stable maintainer:** Already active. Low marginal gain.

Burden evolves as:

$$\text{burden}_{t+1} = \begin{cases} \text{burden}_t + 1 & \text{if } a_t \neq a_0 \\ 0 & \text{if } a_t = a_0 \end{cases}$$

When burden $> B_{\max}$, response probability decays linearly.

Reward. The reward has an immediate and a delayed component.

Immediate reward:

$$R_t^{\text{immediate}} = \alpha \cdot \Delta \text{steps}_t - \beta \cdot \mathbf{1}_{\{a_t \neq a_0\}} + \lambda \cdot \text{goal\_progress}_t$$

Delayed reward (body measures) arrives every 3 weeks ($t = 3k$):

$$R_{3k}^{\text{delayed}} = \eta \cdot \text{BM\_improvement}_{3k}$$

Total reward:

$$R_t = \begin{cases} R_t^{\text{immediate}} + R_t^{\text{delayed}} & t \equiv 0 \pmod 3 \\ R_t^{\text{immediate}} & \text{otherwise} \end{cases}$$

The compound reward (immediate + delayed) is a central research question. Weights $(\alpha, \beta, \lambda, \eta)$ are configurable experiment parameters.

Discount factor. $\gamma \in [0.9, 0.95]$. A high discount factor encourages long-term behaviour change over short-term step increases. Appropriate for health habit formation.

Decision epoch. Daily decisions. This aligns with typical intervention cadence, the 3-week body measure delay (3 epochs), and practical deployment constraints.

4 Framework Design

The framework is organised into six modules that form a pipeline from configuration to results. All extension points use an ABC + registry pattern: implement an abstract class, register it with a key, and reference that key in config — no source code changes required to add new datasets, agents, or behaviour models.

Table 2: Software modules and their dependencies

Module	Role	Depends on
Config	YAML → Pydantic schemas, 3-layer validation	—
Data	Polars lazy reader, FeaturePipeline, StateView	Config
MDP	Environment step/reset, Transition ABC, Reward ABC	Data
UserSim	UserProfile (4 archetypes), ResponseModel ABC	MDP
Agent	Agent ABC, ThompsonSampling (+ optional DQN)	UserSim
Experiment	Factory wiring, CLI, results output	All above

Config files specify dataset paths, MDP parameters, user profile distributions, and agent hyperparameters. The ExperimentFactory reads a composed experiment config, instantiates each component in dependency order, and runs the trial loop. Results are written to CSV with a YAML sidecar for reproducibility. See `docs/code/codebase_plan.md` and `docs/code/code_design.md` for the full specification and `README.md` for a visual dependency chart.

5 Data Sources

This project targets two tiers of data. **Phase 1** uses synthetic data generators parameterised from published population statistics (e.g., NHANES step counts, All of Us sleep distributions). This avoids the 4-8 week lead time required for real dataset access while allowing pipeline development and agent benchmarking.

Phase 2 targets three real datasets for validation: HeartSteps V1/V2 [2, 3] provide MRT data with intervention response mappings for calibrating the user simulator. The All of Us Fitbit Dataset [7] offers 59,000+ participants with wearable data and EHR linkage. The UK Biobank Accelerometer Dataset [8] provides 700,000+ person-days of raw accelerometer data.

Detailed feasibility studies are in `docs/sources/data_sources.md` and `docs/sources/additional_data_sc`

6 Decision Log

Question	Decision	Status
What config format?	YAML	Confirmed
What interface?	CLI + config files; web UI = stretch	Confirmed
Which language and package manager?	Python 3.11 + `uv`	Confirmed
What data for Phase 1?	Synthetic data (real data requires 4-8 weeks)	Confirmed
Which agent first?	Thompson Sampling	Confirmed
How often are decisions made?	Daily epochs	Confirmed
What reward structure?	Multi-timescale: immediate + 3-week delayed	Confirmed
Which user archetypes?	Goal-driven, social, resistant, stable maintainer	Confirmed
How are components wired?	ABC + registry pattern	Confirmed
Which dependencies?	Minimal: no Gymnasium, no SB3	Confirmed
How to distribute?	Both: repo-based now, package if needed later	Confirmed
Is MDP formalisation reasonable?	Awaiting supervisor approval	Open
Which datasets for Phase 2?	HeartSteps first, then All of Us, UK Biobank	Open
What format for experiment output?	CSV, terminal table, JSON, or summary plots	Open
Metrics for success?	Output metrics to evaluate agent performance (regret, reward, adherence)	Open

7 Future Work

Beyond Phase 1, the framework targets LLM-based user simulation using prompt-driven persona modelling, offline RL agents (CQL, IQL) for batch policy learning, non-stationary environment adaptation, and real-data validation against All of Us, UK Biobank, and HeartSteps datasets.

References

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A State Variables

Table 3: State variables

Variable	Type	Description
steps_t	$\mathbb{R}_{\geq 0}$	Step count, past 24h
hr_t	$\mathbb{R}_{\geq 0}$	Heart rate (or resting)
sleep_hours_t	$\mathbb{R}_{\geq 0}$	Previous night sleep
sedentary_min_t	$\mathbb{R}_{\geq 0}$	Sedentary minutes today
time_of_day_t	$\{0, \ldots, 23\}$	Current hour
day_of_week_t	$\{0, \ldots, 6\}$	Day of week
goal_progress_t	$[0, 1]$	Fraction of daily goal met
burden_t	$\mathbb{Z}_{\geq 0}$	Interventions sent today
a_{t-1}	$\mathcal{A}$	Previous action
response_{t-1}	$\{0, 1\}$	Previous response
body_measure_k	$\mathbb{R}$	Clinical measure (q3wks)
age	$\mathbb{Z}_{>0}$	Static profile
gender	$\{0, 1\}$	Static profile
baseline_activity	$\{0, 1, 2\}$	Low / medium / high